

Eukaryotic cells modify RNA after transcription

Enzymes in the eukaryotic nucleus modify pre-mRNA in specific ways before the genetic message is dispatched to the cytoplasm. During this **RNA processing**, both ends of the primary transcript are altered. Also, in most cases, certain interior sections of the RNA molecule are cut out and the remaining parts spliced together. These modifications produce an mRNA molecule ready for translation.

Alteration of mRNA Ends

Each end of a pre-mRNA molecule is modified in a particular way (Figure 17.11). The 3' end, which is synthesized first, receives a **5' cap**, a modified form of a guanine (G) nucleotide added onto the 5' end after transcription of the first 20–40 nucleotides have been transcribed. The 3' end of the pre-mRNA molecule is also modified before the mRNA exits the

nucleus. Recall that the pre-mRNA is cut and released soon after the polyadenylation signal, AAUAAA, is transcribed. At the 3' end, an enzyme then adds 50–250 more adenine (A) nucleotides, forming a **poly-A tail**. The 5' cap and poly-A tail share several important functions. First, they seem to facilitate the export of the mature mRNA from the nucleus. Second, they help protect the mRNA from degradation by hydrolytic enzymes. And third, they help ribosomes attach to the 5' end of the mRNA once it reaches the cytoplasm. In addition to the cap and tail on a eukaryotic mRNA molecule, Figure 17.11 shows the untranslated regions (UTRs) at the 5' and 3' ends of the mRNA (referred to as the 5' UTR and 3' UTR). The UTRs are parts of the mRNA that will not be translated into protein, but they have other functions, such as ribosome binding.

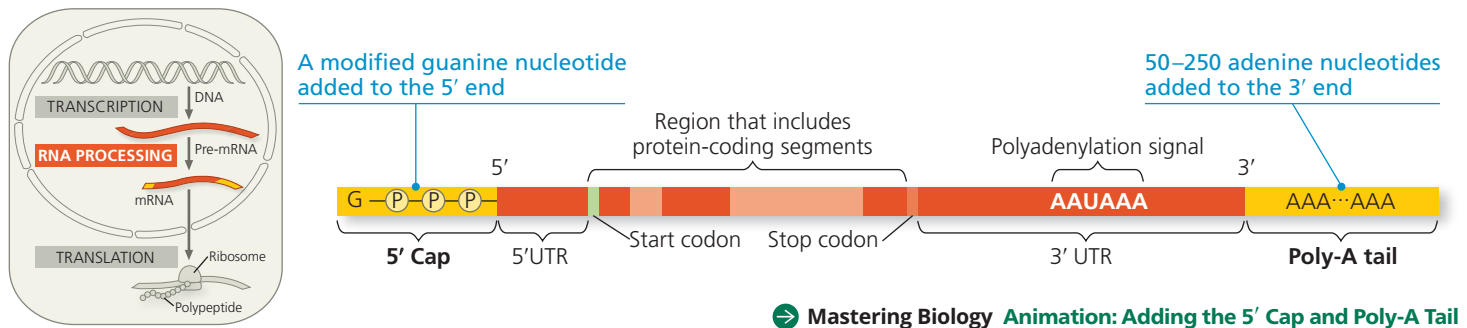
Split Genes and RNA Splicing

A remarkable stage of RNA processing in the eukaryotic nucleus is called **RNA splicing** (Figure 17.12), in which large portions of the RNA primary transcript molecules are removed and the remaining portions are reconnected. This

▼ **Figure 17.11 RNA processing: addition of the 5' cap and poly-A tail.** Enzymes modify the two ends of a eukaryotic pre-mRNA molecule. The modified ends may promote the export of mRNA from the

nucleus, and they help protect the mRNA from degradation. When the mRNA reaches the cytoplasm, the modified ends, in conjunction with certain cytoplasmic proteins, help with ribosome attachment. The 5' cap and

poly-A tail are not translated into protein, nor are the regions called the 5' untranslated region (5' UTR) and 3' untranslated region (3' UTR). The pink segments are introns, which will be described shortly (see Figure 17.12).

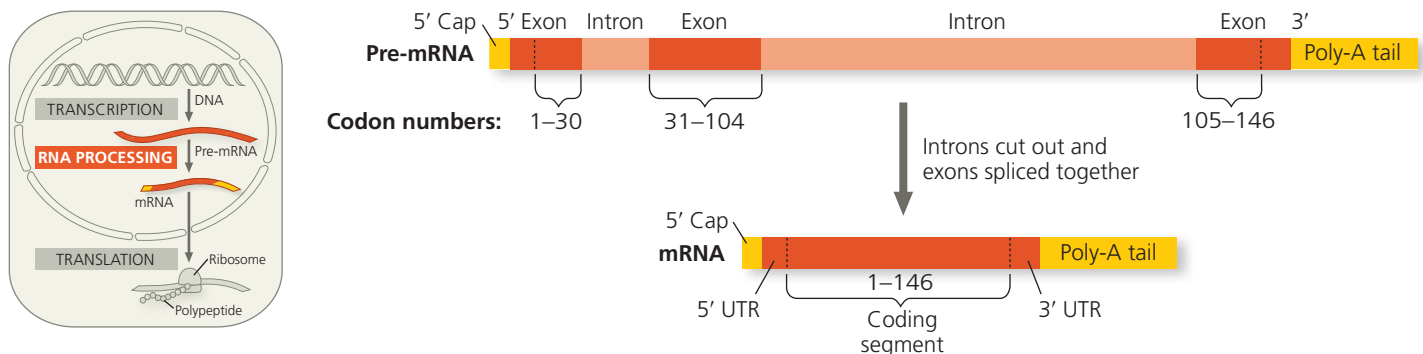


➔ **Mastering Biology Animation: Adding the 5' Cap and Poly-A Tail**

▼ **Figure 17.12 RNA processing: RNA splicing.** The RNA molecule shown here codes for β -globin, one of the polypeptides of hemoglobin. The numbers under the RNA refer to codons; β -globin is 146 amino acids

long. The β -globin gene and its pre-mRNA transcript have three exons, corresponding to sequences that will exit the nucleus as mRNA. (The 5' UTR and 3' UTR do not code for proteins, but they are parts of

exons because they remain in the mRNA.) During RNA processing, the introns are cut out and the exons spliced together. In many genes, the introns are much longer than the exons.



DRAW IT On the mRNA, indicate the sites of the start and stop codons.

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